Department of Chemistry and Biochemistry
University of Arkansas

ORGANIC CUMULATIVE EXAMINATION, 10/27/1990

The following six problems are copied from J. McMurry, "Organic Chemistry" second edition, Brooks/Cole, 1988. (This is the textbook used in our sophomore organic chemistry course.) Do four of these six problems. (Indicate clearly which two are omitted; otherwise we will grade the frist four we come to.

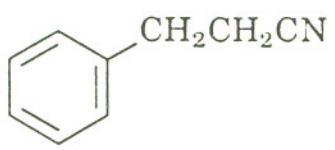
7.25 Compound A has the formula $C_{10}H_{16}$. On catalytic hydrogenation over palladium, it reacts with only 1 equiv of hydrogen. Compound A undergoes reaction with ozone, followed by zinc treatment, to yield a symmetrical diketone, B, with formula $C_{10}H_{16}O_2$.

(a) How many rings does A have? (b) What are the structures of A and B?

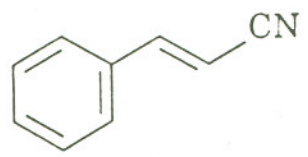
(c) Formulate the reactions.

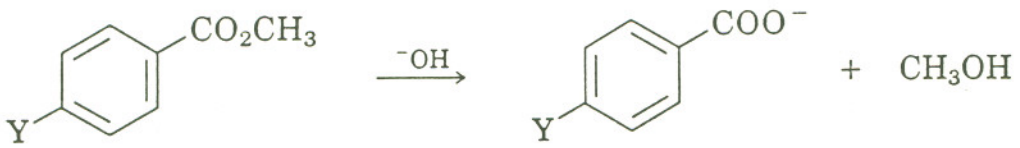
9.54 Compound A, C_7H_{12}, was found to be optically active. On catalytic reduction over a palladium catalyst, 2 equiv of hydrogen were absorbed, yielding compound B, C_7H_{16}. On ozonolysis, two fragments were obtained. One fragment was identified as acetic acid. The other fragment, compound C, was an optically active carboxylic acid, $C_5H_{10}O_2$. Formulate the reactions, and draw structures for compounds A, B, and C.

16.34 How do you explain the fact that electrophilic attack on 3-phenylpropanenitrile occurs at the ortho and para positions, whereas attack on 3-phenylpropenenitrile occurs at the meta position? Use resonance structures of the intermediates in your explanation.



3-Phenylpropanenitrile



3-Phenylpropenenitrile

21.38 In the basic hydrolysis of para-substituted methyl benzoates, the following reactivity order has been found for Y: $NO_2 > Br > H > CH_3 > OCH_3$. How can you explain this reactivity order? Where would you expect Y = C≡N, Y = CHO, and Y = NH_2 to be in the reactivity list?